

(a) Group Rewards

	r_1	r_2	r_3
τ_1	0.1000	0.1800	0.8000
τ_2	0.1800	0.1000	7.2800
τ_3	0.8200	0.9000	8.0000
τ_4	0.9000	0.8200	1.5200
μ	0.5000	0.5000	4.4000
σ	0.4183	0.4183	3.7643
	High Correlation		Large Scale

(b) Covariance Matrix

$$\begin{bmatrix} \sigma_1\sigma_1\rho_{11} = 0.1749 & \sigma_1\sigma_2\rho_{12} = 0.1707 & \sigma_1\sigma_3\rho_{13} = \mathbf{0.1728} \\ \sigma_2\sigma_1\rho_{21} = 0.1707 & \sigma_2\sigma_2\rho_{22} = 0.1749 & \sigma_2\sigma_3\rho_{23} = \mathbf{0.1728} \\ \sigma_3\sigma_1\rho_{31} = \mathbf{0.1728} & \sigma_3\sigma_2\rho_{32} = \mathbf{0.1728} & \sigma_3\sigma_3\rho_{33} = \mathbf{14.1696} \end{bmatrix}$$

$$A_{\text{GRPO}}^i \approx \frac{\Delta R_i}{\sqrt{\mathbf{14.1696} + 2 \times (0.1749 + 0.1707 + 2 \times \mathbf{0.1728})}} \approx \mathbf{0.2536} \Delta R_i$$

(c) Correlation Matrix

$$\begin{bmatrix} \rho_{11} = 1.0000 & \rho_{12} = \mathbf{0.9756} & \rho_{13} = 0.1098 \\ \rho_{21} = \mathbf{0.9756} & \rho_{22} = 1.0000 & \rho_{23} = 0.1098 \\ \rho_{31} = 0.1098 & \rho_{32} = 0.1098 & \rho_{33} = 1.0000 \end{bmatrix}$$

$$A_{\text{CorrGRPO}}^i \approx \frac{\Delta R_i}{\sqrt{3 \times 1.0000 + 2 \times (\mathbf{0.9756} + 2 \times 0.1098)}} \approx \mathbf{0.4307} \Delta R_i$$